

Lab 8

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Question 1: What service appears to be running on port 2200?

No.	Time	Source	Destination	Protocol	Length	Info
32	801.587279	172.31.255.28	10.1.10.33	TCP	60	59632 → 2200 [SYN] Seq=205340159 win=0 Len=0 MSS=1460
33	801.588075	10.1.10.33	172.31.255.28	TCP	60	2200 → 59632 [SYN, ACK] Seq=1127512436 Ack=205340160 win=5840 Len=0 MSS=1460
40	804.425254	172.31.255.28	10.1.10.33	TCP	60	59632 → 2200 [RST] Seq=205340160 win=0 Len=0
42	804.876700	172.31.255.28	10.1.10.33	TCP	60	59643 → 2200 [SYN] Seq=2078117135 win=0 Len=0 MSS=1460
43	804.878706	10.1.10.33	172.31.255.28	TCP	60	2200 → 59643 [SYN, ACK] Seq=2230794879 Ack=2078117135 win=5840 Len=0 MSS=1460
46	804.468457	172.31.255.28	10.1.10.33	TCP	60	59643 → 2200 [RST] Seq=2078117135 win=0 Len=0
52	838.409960	172.31.255.28	10.1.10.33	TCP	60	62987 → 2200 [SYN] Seq=1069715935 win=0 Len=0 MSS=1460
53	838.409189	10.1.10.33	172.31.255.28	TCP	60	2200 → 62987 [SYN, ACK] Seq=106974568 Ack=1097159260 win=5840 Len=0 MSS=1460
54	839.574849	172.31.255.28	10.1.10.33	TCP	60	62987 → 2200 [RST] Seq=1097159260 win=0 Len=0
61	841.603356	172.31.255.28	10.1.10.33	TCP	60	62987 → 2200 [SYN] Seq=106974568 Ack=1097159260 win=5840 Len=0 MSS=1460
62	841.628454	10.1.10.33	172.31.255.28	TCP	60	2200 → 62987 [SYN, ACK] Seq=1342738887 Ack=106974568 win=5840 Len=0 MSS=1460
66	841.702369	172.31.255.28	10.1.10.33	TCP	60	62987 → 2200 [RST] Seq=106974568 Ack=1342738887 win=0 Len=0
68	842.983872	172.31.255.28	10.1.10.33	TCP	60	62987 → 2200 [SYN] Seq=1330713379 win=0 Len=0 MSS=1460
69	842.954330	10.1.10.33	172.31.255.28	TCP	60	2200 → 62987 [SYN, ACK] Seq=2312494004 Ack=1330713379 win=5840 Len=0 MSS=1460
70	843.014462	172.31.255.28	10.1.10.33	TCP	60	62987 → 2200 [RST] Seq=1330713379 win=0 Len=0
71	958.513357	172.31.255.28	10.1.10.33	TCP	74	40679 → 2200 [SYN] Seq=3861212989 Ack=3861212991 win=5888 Len=0 SACK_PERM TSval=129728427 TSecr=1671690274 US=64
72	958.512696	10.1.10.33	172.31.255.28	TCP	74	2200 → 40679 [SYN, ACK] Seq=722104187 Ack=3861212989 win=5792 Len=0 SACK_PERM TSval=129728427 TSecr=1671690274 US=32
73	958.574508	172.31.255.28	10.1.10.33	TCP	66	40679 → 2200 [ACK] Seq=3861212989 Ack=722104187 win=5888 Len=0 TSval=1671690289 TSecr=129728427
74	958.575209	10.1.10.33	172.31.255.28	TCP	66	2200 → 40679 [ACK] Seq=722104187 Ack=3861212989 win=5888 Len=0 TSval=1671690289 TSecr=129728427
75	958.572209	172.31.255.28	10.1.10.33	TCP	66	40679 → 2200 [ACK] Seq=3861212989 Ack=722104187 win=5888 Len=0 TSval=1671690289 TSecr=129728427
80	959.971164	172.31.255.28	10.1.10.33	TCP	66	40679 → 2200 [PSH, ACK] Seq=3861212989 Ack=722104187 win=5888 Len=2 TSval=1671690640 TSecr=129728452
81	959.972733	10.1.10.33	172.31.255.28	TCP	66	2200 → 40679 [ACK] Seq=722104187 Ack=3861212991 win=5792 Len=0 TSval=129728791 TSecr=1671690640
82	959.972738	10.1.10.33	172.31.255.28	TCP	85	2200 → 40679 [PSH, ACK] Seq=722104187 Ack=3861212991 win=5792 Len=19 TSval=129728791 TSecr=1671690640
83	959.972740	10.1.10.33	172.31.255.28	TCP	66	2200 → 40679 [FIN, ACK] Seq=722104206 Ack=3861212991 win=5792 Len=0 TSval=129728791 TSecr=1671690640
84	960.832963	172.31.255.28	10.1.10.33	TCP	66	40679 → 2200 [ACK] Seq=3861212991 Ack=722104206 win=5888 Len=0 TSval=1671690656 TSecr=129728791
85	960.834666	172.31.255.28	10.1.10.33	TCP	66	40679 → 2200 [FIN, ACK] Seq=3861212991 Ack=722104207 win=5888 Len=0 TSval=1671690656 TSecr=129728791
86	960.835511	10.1.10.33	172.31.255.28	TCP	66	2200 → 40679 [ACK] Seq=722104207 Ack=3861212992 win=5792 Len=0 TSval=129728867 TSecr=1671690656
152	1089.738568	172.31.255.28	10.1.10.33	TCP	74	40683 → 2200 [SYN] Seq=1645130522 Ack=164674464 win=5792 Len=0 SACK_PERM TSval=1671723867 TSecr=0 US=64
153	1089.739644	10.1.10.33	172.31.255.28	TCP	74	2200 → 40683 [SYN, ACK] Seq=1645130522 Ack=164674464 win=5792 Len=0 SACK_PERM TSval=129751219 TSecr=1671723867 US=32

Using “entry.pcap”, I was able to apply the “tcp.port == 2200” display filter, which would only show me all relevant packets. I chose to inspect frame 74 more closely.

Checksum: 0x7df5 [unverified]	0000	00000000	00100110	11110011	00110000	00011111	11100010	00001000	00000000	& 0
[Checksum Status: Unverified]	0006	00100111	11111011	10110000	00010000	00010000	00000000	01000101	00000000	E
Urgent Pointer: 0	0010	00000000	01010101	00010000	11010110	01000000	00000000	01000000	00000110	@
Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps	0016	01010101	01010001	00001010	00000001	00001010	00100001	10101100	00011111	jq
TCP Option - No-Operation (NOP)	0020	11111111	00011100	00001000	10011000	10011110	11000111	00101011	00001010	+
Kind: No-Operation (1)	0026	01011111	01010100	11000110	00100101	01011101	00111011	10000000	00011000	oT
TCP Option - No-Operation (NOP)	0030	00000000	10110101	01111011	11110101	00000000	00000000	00000001	00000001	..)
Kind: No-Operation (1)	0036	00001000	00001010	00000111	10110111	01111111	11000100	01100011	10100011	6
TCP Option - Timestamps: TSval 129728452, TSecr 1671690289	0040	11111000	00100001	01010011	01010011	01010000	00101001	00100010	00101110	..SSHT-
Kind: Time Stamp Option (8)	0046	00100000	00101101	01001111	01100000	01001011	01011010	01010011	01010011	0-OpenS
Length: 10	0050	01001000	01011111	00101101	00101110	00110000	00110000	00100000	00100000	H_5_3p1
Timestamp value: 129728452	0056	01000100	01000101	01000010	01010001	01000001	01101110	00101001	00110011	Debian-
Timestamp echo reply: 1671690289	0060	01110101	01100010	01110101	01011110	01110100	01101001	00011011	00001101	ubuntu7
	0066	00001010								.
[Timestamps]										
[Time since first frame in this TCP stream: 0.102634000 seconds]										
[Time since previous frame in this TCP stream: 0.039393000 seconds]										
[SEQ/ACK analysis]										
[RTT: 0.063241000 seconds]										
[Bytes in flight: 39]										
[Bytes sent since last PSH flag: 39]										

```
1SSH-2.
0-OpenSS
H_5.3p1
Debian-3
ubuntu7.
```

“TCP payload” caught my attention so I clicked on it to view its respective metadata. Here, I was able to confirm that the service running on port 2200 was **Secure Shell**, which is interesting since the default SSH port is 22.

Question 3: Which IP address had port 2200 open?

32	801.587279	172.31.255.28	10.1.10.33	TCP	60	59632 → 2200	[SYN]	Seq=2053340159 Win=2048 Len=0 MSS=1460
33	801.588075	10.1.10.33	172.31.255.28	TCP	60	2200 → 59632	[SYN, ACK]	Seq=1127512436 Ack=2053340160 Win=5840 Len=0 MSS=1460
40	801.643654	172.31.255.28	10.1.10.33	TCP	60	59632 → 2200	[RST]	Seq=2053340160 Win=0 Len=0
42	804.077670	172.31.255.28	10.1.10.33	TCP	60	59643 → 2200	[SYN]	Seq=2070117119 Win=2048 Len=0 MSS=1460
43	804.078786	10.1.10.33	172.31.255.28	TCP	60	2200 → 59643	[SYN, ACK]	Seq=2230794079 Ack=2070117120 Win=5840 Len=0 MSS=1460
46	804.148647	172.31.255.28	10.1.10.33	TCP	60	59643 → 2200	[RST]	Seq=2070117120 Win=0 Len=0
52	838.490909	172.31.255.28	10.1.10.33	TCP	60	62987 → 2200	[SYN]	Seq=1097159259 Win=3072 Len=0 MSS=1460
53	838.491869	10.1.10.33	172.31.255.28	TCP	60	2200 → 62987	[SYN, ACK]	Seq=1506974988 Ack=1097159260 Win=5840 Len=0 MSS=1460
54	838.574849	172.31.255.28	10.1.10.33	TCP	60	62987 → 2200	[RST]	Seq=1097159260 Win=0 Len=0
61	841.620326	172.31.255.28	10.1.10.33	TCP	60	62998 → 2200	[SYN]	Seq=1080382299 Win=3072 Len=0 MSS=1460
62	841.620454	10.1.10.33	172.31.255.28	TCP	60	2200 → 62998	[SYN, ACK]	Seq=1342738007 Ack=1080382300 Win=5840 Len=0 MSS=1460
66	841.702399	172.31.255.28	10.1.10.33	TCP	60	62998 → 2200	[RST]	Seq=1080382300 Win=0 Len=0
68	842.953872	172.31.255.28	10.1.10.33	TCP	60	62999 → 2200	[SYN]	Seq=1130713179 Win=4096 Len=0 MSS=1460
69	842.954330	10.1.10.33	172.31.255.28	TCP	60	2200 → 62999	[SYN, ACK]	Seq=2312494904 Ack=1130713180 Win=5840 Len=0 MSS=1460
70	843.014462	172.31.255.28	10.1.10.33	TCP	60	62999 → 2200	[RST]	Seq=1130713180 Win=0 Len=0

Using “entry.pcap” with the same filter (tcp.port == 2200), I was able to tell that most of the transmitted “SYN” packets didn’t get a response which is evident by the “RST” packets.

71	958.511357	172.31.255.28	10.1.10.33	TCP	74	40679 → 2200	[SYN]	Seq=3861212988 Win=5840 Len=0 MSS=1460 SACK_PERM TSval=1671690274 TSecr=0 WS=64
72	958.512696	10.1.10.33	172.31.255.28	TCP	74	2200 → 40679	[SYN, ACK]	Seq=722104147 Ack=3861212989 Win=5792 Len=0 MSS=1460 SACK_PERM TSval=129728427
73	958.574598	172.31.255.28	10.1.10.33	TCP	66	40679 → 2200	[ACK]	Seq=3861212989 Ack=722104148 Win=5888 Len=0 TSval=1671690289 TSecr=129728427
74	958.613991	10.1.10.33	172.31.255.28	TCP	105	2200 → 40679	[PSH, ACK]	Seq=722104148 Ack=3861212989 Win=5792 Len=39 TSval=129728452 TSecr=1671690289
75	958.672209	172.31.255.28	10.1.10.33	TCP	66	40679 → 2200	[ACK]	Seq=3861212989 Ack=722104187 Win=5888 Len=0 TSval=1671690316 TSecr=129728452
80	959.971164	172.31.255.28	10.1.10.33	TCP	68	40679 → 2200	[PSH, ACK]	Seq=3861212989 Ack=722104187 Win=5888 Len=2 TSval=1671690640 TSecr=129728452

From frames 71-73, I can see that there was finally an “ACK” from IP address 10.1.10.33, confirming that port 2200 is open.

Question 4: Which version of SSH was the attacker using?

No.	Time	Source	Destination	Protocol	Length	Info
139	1028.880307	10.1.10.60	172.31.255.28	SSH	105	Server: Protocol (SSH-2.0-OpenSSH_5.1p1 Debian-3ubuntu1)
141	1031.362592	172.31.255.28	10.1.10.60	SSH	68	Client: Encrypted packet (len=2)
143	1031.363600	10.1.10.60	172.31.255.28	SSH	85	Server: Encrypted packet (len=19)

```
1000 .... = Header Length: 32 bytes (8)
  Flags: 0x018 (PSH, ACK)
  Window: 91
  [Calculated window size: 5824]
  [Window size scaling factor: 64]
  Checksum: 0xd70 [unverified]
  [Checksum Status: Unverified]
  Urgent Pointer: 0
  Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps
  [Timestamps]
  [SEQ/ACK analysis]
  TCP payload (39 bytes)
  SSH Protocol
    Protocol: SSH-2.0-OpenSSH_5.1p1 Debian-3ubuntu1
    [Direction: server-to-client]
```

On the same file, I was able to use “ssh” as a filter. Luckily, the “Info” section of the frame offered the answer. The version of SSH that the attacker was using was **OpenSSH 5.1p1**.

Question 6: How many IP addresses had port 3389 open?

tcp.flags.syn == 1 && tcp.flags.ack == 1 && tcp.port == 3389					
No.	Time	Source	Destination	Protocol	Length Info
1012	25.792990	10.1.10.20	10.1.10.33	TCP	74 3389 → 46009 [SYN, ACK] Seq=1188584703 Ack=34564360 Win=8192 Len=0 MSS=1460 WS=256 SACK_PERM TSval=55798963 TSecr=131203382

```
Frame 1012: 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface 0
Ethernet II, Src: Dell_ab:23:be (14:fe:b5:ab:23:be), Dst: PCSSystemtec_fb:b8:10 (08:00:27:fb:b8:10)
Internet Protocol Version 4, Src: 10.1.10.20, Dst: 10.1.10.33
Transmission Control Protocol, Src Port: 3389, Dst Port: 46009, Seq: 1188584703, Ack: 34564360, Len: 0
  Source Port: 3389
  Destination Port: 46009
  [Stream index: 48]
  [Stream Packet Number: 2]
  [Conversation completeness: Complete, NO_DATA (39)]
  ...1... = RST: Present
  ...0... = FIN: Absent
  ...0... = Data: Absent
  ...1... = ACK: Present
  ...1... = SYN-ACK: Present
  ...1... = SYN: Present
  [Completeness Flags: R-ASS]
  [TCP Segment Len: 0]
  Sequence Number: 1188584703
  [Next Sequence Number: 1188584704]
  Acknowledgment Number: 34564360
  1010 ... = Header Length: 40 bytes (10)
  Flags: 0x012 (SYN, ACK)
  Window: 8192
  [Calculated window size: 8192]
```

Using “init.recon.pcap” and applying the filter “tcp.flags.syn == 1 && tcp.flags.ack == 1 && tcp.port == 3389”, I determined that only **one** IP address had port 3389 open. The filter shows only SYN-ACK packets sent from port 3389, which indicate that the IP sending these packets is listening on that port. Based on this, I found that 10.1.10.20 was sending a SYN-ACK packet from port 3389, confirming that it had port 3389 open.

Question 7: Which of the following IP addresses appears to be running a Human Machine Interface (HMI)?

I began filtering by the given IP addresses using “ip.src == ...”.

ip.src == 10.1.10.60					
Time	Source	Destination	Protocol	Length	Info
367	23.640544	10.1.10.60	10.1.10.33	TCP	54 80 → 36189 [RST, ACK] Seq=0 Ack=2854746301 Win=0 Len=0
942	25.779225	10.1.10.60	10.1.10.33	TCP	54 443 → 36455 [RST, ACK] Seq=0 Ack=2658918134 Win=0 Len=0
983	25.788214	10.1.10.60	10.1.10.33	TCP	54 21 → 35514 [RST, ACK] Seq=0 Ack=65880941 Win=0 Len=0
1017	25.793540	10.1.10.60	10.1.10.33	TCP	54 80 → 36988 [RST, ACK] Seq=0 Ack=1644099652 Win=0 Len=0
1030	25.797749	10.1.10.60	10.1.10.33	TCP	54 3389 → 47139 [RST, ACK] Seq=0 Ack=823520183 Win=0 Len=0
1055	25.801426	10.1.10.60	10.1.10.33	TCP	74 22 → 56140 [SYN, ACK] Seq=1395014073 Ack=3531508851 Win=0
1061	25.801499	10.1.10.60	10.1.10.33	TCP	54 23 → 42013 [RST, ACK] Seq=0 Ack=3162002567 Win=0 Len=0

ip.src == 10.1.10.130					
Time	Source	Destination	Protocol	Length	Info
222	23.317784	10.1.10.130	10.1.10.33	TCP	62 80 → 51563 [SYN, ACK] Seq=599828439 Ack=166792409 Win=2048 Len=0 MSS=16384 SACK_PERM
1044	25.797856	10.1.10.130	10.1.10.33	TCP	62 80 → 52491 [SYN, ACK] Seq=599913876 Ack=1856340824 Win=2048 Len=0 MSS=16384 SACK_PERM
1589	205.786769	10.1.10.130	10.1.10.33	TCP	62 80 → 52537 [SYN, ACK] Seq=599984203 Ack=3643019798 Win=2048 Len=0 MSS=16384 SACK_PERM
1592	205.860749	10.1.10.130	10.1.10.33	TCP	159 80 → 52537 [PSH, ACK] Seq=599984204 Ack=3643019931 Win=2048 Len=105 [TCP PDU reassembled in 1596]
1594	205.872470	10.1.10.130	10.1.10.33	TCP	895 80 → 52537 [PSH, ACK] Seq=599984309 Ack=3643019931 Win=2048 Len=841 [TCP PDU reassembled in 1596]
1596	205.880651	10.1.10.130	10.1.10.33	HTTP	60 HTTP/1.0 200 OK (text/html)
1598	205.882655	10.1.10.130	10.1.10.33	TCP	60 80 → 52537 [RST] Seq=599985151 Win=0 Len=0
1655	243.776189	10.1.10.130	10.1.10.33	TCP	62 80 → 52538 [SYN, ACK] Seq=600062339 Ack=1564897829 Win=2048 Len=0 MSS=16384 SACK_PERM
1658	243.839278	10.1.10.130	10.1.10.33	TCP	159 80 → 52538 [PSH, ACK] Seq=600062340 Ack=1564897962 Win=2048 Len=105 [TCP PDU reassembled in 1662]
1660	243.850568	10.1.10.130	10.1.10.33	TCP	895 80 → 52538 [PSH, ACK] Seq=600062445 Ack=1564897962 Win=2048 Len=841 [TCP PDU reassembled in 1662]
1662	243.859090	10.1.10.130	10.1.10.33	HTTP	60 HTTP/1.0 200 OK (text/html)
1664	243.860265	10.1.10.130	10.1.10.33	TCP	60 80 → 52538 [RST] Seq=600062387 Win=0 Len=0

The first two, “10.1.10.60” and “10.1.10.130” didn’t stand out much, besides the HTTP packet from “10.1.10.130” which didn’t yield anything.

ip.src == 10.1.10.29					
Time	Source	Destination	Protocol	Length	Info
152 23.006482	10.1.10.29	10.1.10.33	TCP	74	80 → 50204 [SYN, ACK] Seq=1287653980 Ack=2797296464 Win=16384 Len=0 MSS=1460 WS=1 TSval=3733420 TSecr=131202684
947 25.779942	10.1.10.29	10.1.10.33	TCP	74	443 → 33477 [SYN, ACK] Seq=1383517987 Ack=1008617757 Win=16384 Len=0 MSS=1460 WS=1 TSval=3733425 TSecr=131203378
964 25.783463	10.1.10.29	10.1.10.33	TCP	74	21 → 33164 [SYN, ACK] Seq=1401932692 Ack=948412402 Win=16384 Len=0 MSS=1460 WS=1 TSval=3733425 TSecr=131203379
987 25.789589	10.1.10.29	10.1.10.33	TCP	74	80 → 51194 [ACK] Seq=14096090941 Ack=2741130964 Win=16384 Len=0 MSS=1460 WS=1 TSval=3733425 TSecr=131203380
1030 25.797006	10.1.10.29	10.1.10.33	TCP	68	3309 → 49390 [RST, ACK] Seq=90 Ack=2446693188 Win=0 Len=0
1065 25.801536	10.1.10.29	10.1.10.33	TCP	68	22 → 48026 [RST, ACK] Seq=0 Ack=3931161741 Win=0 Len=0
1072 25.801653	10.1.10.29	10.1.10.33	TCP	74	23 → 46799 [SYN, ACK] Seq=1449782032 Ack=2304587552 Win=16384 Len=0 MSS=1460 WS=1 TSval=3733426 TSecr=131203383
1399 167.830759	10.1.10.29	10.1.10.33	TCP	74	80 → 51251 [SYN, ACK] Seq=1917489210 Ack=3807823749 Win=16384 Len=0 MSS=1460 WS=1 TSval=3733709 TSecr=131238866
1402 167.872144	10.1.10.29	10.1.10.33	TCP	68	80 → 51251 [ACK] Seq=1917489211 Ack=3807823881 Win=17368 Len=0 TSval=3733709 TSecr=131238877
1403 167.887039	10.1.10.29	10.1.10.33	TCP	352	80 → 51251 [PSH, ACK] Seq=1917489211 Ack=3807823881 Win=17520 Len=286 TSval=3733710 TSecr=131238877 [TCP PDU reassembled in 1405]
1405 167.887085	10.1.10.29	10.1.10.33	HTTP	606	HTTP/1.0 200 OK (text/html)
1407 167.889439	10.1.10.29	10.1.10.33	TCP	68	80 → 51251 [FIN, ACK] Seq=1917490127 Ack=3807823881 Win=17520 Len=0 TSval=3733710 TSecr=131238877
1409 167.898775	10.1.10.29	10.1.10.33	TCP	68	80 → 51251 [ACK] Seq=1917490128 Ack=3807823882 Win=17520 Len=0 TSval=3733710 TSecr=131238883
1467 183.743351	10.1.10.29	10.1.10.33	TCP	74	23 → 46823 [SYN, ACK] Seq=2474065530 Ack=1424875672 Win=16384 Len=0 MSS=1460 WS=1 TSval=3733741 TSecr=131242842
1470 183.744102	10.1.10.29	10.1.10.33	TCP	66	23 → 46823 [ACK] Seq=2474065531 Ack=1424875696 Win=17496 Len=0 TSval=3733741 TSecr=131242842
1474 183.826386	10.1.10.29	10.1.10.33	TELNET	72	Will Echo, Will Suppress Go Ahead
1477 183.832091	10.1.10.29	10.1.10.33	TCP	66	23 → 46823 [ACK] Seq=2474065537 Ack=1424875699 Win=17517 Len=0 TSval=3733741 TSecr=131242863
1478 183.832906	10.1.10.29	10.1.10.33	TELNET	129	63 bytes data
1484 185.549982	10.1.10.29	10.1.10.33	TCP	66	23 → 46823 [ACK] Seq=2474065600 Ack=1424875701 Win=17518 Len=0 TSval=3733745 TSecr=131243293
1485 185.550996	10.1.10.29	10.1.10.33	TELNET	68	2 bytes data
1487 185.551938	10.1.10.29	10.1.10.33	TELNET	76	10 bytes data
1493 186.583891	10.1.10.29	10.1.10.33	TCP	66	23 → 46823 [ACK] Seq=2474065612 Ack=1424875703 Win=17518 Len=0 TSval=3733747 TSecr=131243531
1494 186.517494	10.1.10.29	10.1.10.33	TELNET	93	27 bytes data
1500 187.242623	10.1.10.29	10.1.10.33	TCP	66	23 → 46823 [ACK] Seq=2474065639 Ack=1424875705 Win=17518 Len=0 TSval=3733748 TSecr=131243715
1501 187.242630	10.1.10.29	10.1.10.33	TELNET	68	2 bytes data
1504 187.245194	10.1.10.29	10.1.10.33	TELNET	76	10 bytes data
1511 187.437116	10.1.10.29	10.1.10.33	TCP	66	23 → 46823 [ACK] Seq=2474065651 Ack=1424875707 Win=17518 Len=0 TSval=3733749 TSecr=131243765
1516 188.345188	10.1.10.29	10.1.10.33	TCP	66	23 → 46823 [ACK] Seq=2474065651 Ack=1424875709 Win=17518 Len=0 TSval=3733750 TSecr=131243991
1517 189.461522	10.1.10.29	10.1.10.33	TELNET	93	27 bytes data
1519 189.463011	10.1.10.29	10.1.10.33	TELNET	78	12 bytes data
1526 190.795594	10.1.10.29	10.1.10.33	TCP	66	23 → 46823 [ACK] Seq=2474065690 Ack=1424875711 Win=17518 Len=0 TSval=3733755 TSecr=131244604
1527 190.814068	10.1.10.29	10.1.10.33	TELNET	104	30 bytes data
1528 190.814083	10.1.10.29	10.1.10.33	TCP	66	23 → 46823 [FIN, ACK] Seq=2474065728 Ack=1424875711 Win=17520 Len=0 TSval=3733755 TSecr=131244604
1532 190.824645	10.1.10.29	10.1.10.33	TCP	66	23 → 46823 [ACK] Seq=2474065729 Ack=1424875712 Win=17520 Len=0 TSval=3733755 TSecr=131244609

Then, “10.1.10.29” seemed more interesting, but after inspecting the HTTP and TELNET packets, there was evidence of it running a HMI.

ip.src == 10.1.10.20					
Time	Source	Destination	Protocol	Length	Info
378 23.641193	10.1.10.20	10.1.10.33	TCP	74	80 → 43356 [SYN, ACK] Seq=1453656426 Ack=3936921650 Win=0 Len=0 MSS=1460 WS=256 SACK_PERM TSval=5578740 TSecr=131202842
655 25.772328	10.1.10.20	10.1.10.33	TCP	74	80 → 44216 [SYN, ACK] Seq=1453656426 Ack=3936921650 Win=0 Len=0 MSS=1460 WS=256 SACK_PERM TSval=5578740 TSecr=131202842
656 25.782239	10.1.10.20	10.1.10.33	TCP	68	21 → 48813 [RST, ACK] Seq=0 Ack=9203184 Win=0 Len=0
675 25.787528	10.1.10.20	10.1.10.33	TCP	74	80 → 44133 [SYN, ACK] Seq=348954400 Ack=140477321 Win=0 Len=0 MSS=1460 WS=256 SACK_PERM TSval=5578962 TSecr=131203188
1012 25.790968	10.1.10.20	10.1.10.33	TCP	74	100 → 44044 [SYN, ACK] Seq=3188084703 Ack=1404444444 Win=0 Len=0 MSS=1460 WS=256 SACK_PERM TSval=5578968 TSecr=131203192
1026 25.799227	10.1.10.20	10.1.10.33	TCP	68	22 → 34252 [RST, ACK] Seq=0 Ack=320383707 Win=0 Len=0
1031 25.799255	10.1.10.20	10.1.10.33	TCP	68	21 → 10851 [RST, ACK] Seq=0 Ack=320383707 Win=0 Len=0
1331 140.951133	10.1.10.20	10.1.10.33	TCP	74	80 → 44218 [SYN, ACK] Seq=1267606922 Ack=1134820608 Win=0 Len=0 MSS=1460 WS=256 SACK_PERM TSval=55810433 TSecr=131232851
1334 140.950107	10.1.10.20	10.1.10.33	HTTP	466	HTTP/1.1 401 Access Denied (text/html)
1337 140.950242	10.1.10.20	10.1.10.33	TCP	66	80 → 44218 [ACK] Seq=1267607031 Ack=113482193 Win=0 Len=0 MSS=1460 WS=256 SACK_PERM TSval=55810438 TSecr=131232863
1338 140.950244	10.1.10.20	10.1.10.33	TCP	66	80 → 44218 [ACK] Seq=1267607031 Ack=113482193 Win=0 Len=0 MSS=1460 WS=256 SACK_PERM TSval=55810438 TSecr=131232863
1348 278.337219	10.1.10.20	10.1.10.33	TCP	74	80 → 44218 [SYN, ACK] Seq=3773696319 Ack=413363560 Win=0 Len=0 MSS=1460 WS=256 SACK_PERM TSval=55824195 TSecr=131266449
1777 278.283409	10.1.10.20	10.1.10.33	HTTP	466	HTTP/1.1 401 Access Denied (text/html)
1710 278.293226	10.1.10.20	10.1.10.33	TCP	66	80 → 44216 [ACK] Seq=3773696434 Ack=413363563 Win=0 Len=0 MSS=1460 WS=256 SACK_PERM TSval=55824201 TSecr=131266463
1711 278.293229	10.1.10.20	10.1.10.33	TCP	66	80 → 44216 [FIN, ACK] Seq=3773696434 Ack=413363563 Win=0 Len=0 MSS=1460 WS=256 SACK_PERM TSval=55824201 TSecr=131266463
1751 282.217486	10.1.10.20	10.1.10.33	TCP	74	80 → 44217 [SYN, ACK] Seq=3299520948 Ack=4080598949 Win=0 Len=0 MSS=1460 WS=256 SACK_PERM TSval=55824644 TSecr=131267568
1754 282.218083	10.1.10.20	10.1.10.33	HTTP	425	HTTP/1.1 401 Access Denied (text/html)
1757 282.223807	10.1.10.20	10.1.10.33	TCP	66	80 → 44217 [ACK] Seq=3299520948 Ack=4080598949 Win=0 Len=0 MSS=1460 WS=256 SACK_PERM TSval=55824644 TSecr=131267569
1758 282.223809	10.1.10.20	10.1.10.33	TCP	66	80 → 44217 [FIN, ACK] Seq=3299520948 Ack=4080598949 Win=0 Len=0 MSS=1460 WS=256 SACK_PERM TSval=55824644 TSecr=131267569
1801 287.125768	10.1.10.20	10.1.10.33	TCP	74	80 → 44218 [SYN, ACK] Seq=4157776317 Ack=3138430849 Win=0 Len=0 MSS=1460 WS=256 SACK_PERM TSval=55825084 TSecr=131268670
1804 287.125771	10.1.10.20	10.1.10.33	HTTP	425	HTTP/1.1 401 Access Denied (text/html)
1807 287.125962	10.1.10.20	10.1.10.33	TCP	66	80 → 44218 [ACK] Seq=4157776317 Ack=3138430849 Win=0 Len=0 MSS=1460 WS=256 SACK_PERM TSval=55825085 TSecr=131268671
1808 287.125964	10.1.10.20	10.1.10.33	TCP	66	80 → 44218 [FIN, ACK] Seq=4157776317 Ack=3138430849 Win=0 Len=0 MSS=1460 WS=256 SACK_PERM TSval=55825085 TSecr=131268671
1835 289.745056	10.1.10.20	10.1.10.33	TCP	74	80 → 44219 [SYN, ACK] Seq=1608764011 Ack=2418123613 Win=0 Len=0 MSS=1460 WS=256 SACK_PERM TSval=55825347 TSecr=131269320
1838 289.757840	10.1.10.20	10.1.10.33	HTTP	425	HTTP/1.1 401 Access Denied (text/html)
1841 289.760911	10.1.10.20	10.1.10.33	TCP	66	80 → 44219 [ACK] Seq=1608764371 Ack=2418123613 Win=0 Len=0 MSS=1460 WS=256 SACK_PERM TSval=55825347 TSecr=131269328
1842 289.760933	10.1.10.20	10.1.10.33	TCP	66	80 → 44219 [FIN, ACK] Seq=1608764371 Ack=2418123613 Win=0 Len=0 MSS=1460 WS=256 SACK_PERM TSval=55825347 TSecr=131269328
1862 295.596939	10.1.10.20	10.1.10.33	TCP	74	80 → 44218 [SYN, ACK] Seq=2722281812 Ack=912118459 Win=0 Len=0 MSS=1460 WS=256 SACK_PERM TSval=55825926 TSecr=131270776
1865 295.596945	10.1.10.20	10.1.10.33	TCP	66	80 → 44218 [FIN, ACK] Seq=2722281812 Ack=912118459 Win=0 Len=0 MSS=1460 WS=256 SACK_PERM TSval=55825927 TSecr=131270777
1895 299.582680	10.1.10.20	10.1.10.33	TCP	74	80 → 44221 [SYN, ACK] Seq=2470196493 Ack=2812807979 Win=0 Len=0 MSS=1460 WS=256 SACK_PERM TSval=55826329 TSecr=131271781
1900 299.583773	10.1.10.20	10.1.10.33	HTTP	425	HTTP/1.1 401 Access Denied (text/html)
1901 299.588908	10.1.10.20	10.1.10.33	TCP	66	80 → 44221 [ACK] Seq=2470196493 Ack=2812807979 Win=0 Len=0 MSS=1460 WS=256 SACK_PERM TSval=55826330 TSecr=131271783
1902 299.588918	10.1.10.20	10.1.10.33	TCP	66	80 → 44221 [FIN, ACK] Seq=2470196493 Ack=2812807979 Win=0 Len=0 MSS=1460 WS=256 SACK_PERM TSval=55826330 TSecr=131271783

```
▼ Hypertext Transfer Protocol
  ▼ HTTP/1.1 401 Access Denied\r\n
    Response Version: HTTP/1.1
    Status Code: 401
    [Status Code Description: Unauthorized]
    Response Phrase: Access Denied
    WWW-Authenticate: Digest realm="PeakHMI",qop="auth",
    WWW-Authenticate: Basic Realm="PeakHMI"\r\n
    Content-Type: text/html\r\n
```

Finally, when filtering for “10.1.10.20”, I noticed some interesting HTTP packets with “Access Denied”. After looking at it closer, I found “PeakHMI” which suggests “10.1.10.20” is the IP address running a HMI.

Question 11: Which of the following best describes the nature of the communications between .20 and .130?

Time	Source	Destination	Protocol	Length	Info
3 0.002152	10.1.10.20	10.1.10.130	TCP	66	49242 → 44818 [SYN] Seq=1242259998 Win=8192 Len=0 MSS=1460 WS=256 SACK_PERM
6 0.008337	10.1.10.130	10.1.10.20	TCP	62	44818 → 49242 [SYN, ACK] Seq=2337482508 Ack=1242259999 Win=2000 Len=0 MSS=16384 SACK_PERM
7 0.008344	10.1.10.20	10.1.10.130	TCP	60	49242 → 44818 [ACK] Seq=1242259999 Ack=2337482509 Win=64240 Len=0
8 0.229368	10.1.10.20	10.1.10.130	ENIP	82	Register Session (Req), Session: 0x00000000
9 0.237784	10.1.10.130	10.1.10.20	ENIP	82	Register Session (Resp), Session: 0x02A7A67A
10 0.245076	10.1.10.20	10.1.10.130	CIP CH	140	Connection Manager - Forward Open (Message Router)
11 0.250004	10.1.10.130	10.1.10.20	CIP CH	124	Success: Connection Manager - Forward Open (Message Router)
12 0.256118	10.1.10.20	10.1.10.130	CIP PC	127	PCCC Class - Execute PCCC
13 0.265889	10.1.10.130	10.1.10.20	CIP PC	117	Success: PCCC Class - Execute PCCC
14 0.467822	10.1.10.20	10.1.10.130	TCP	60	49242 → 44818 [ACK] Seq=1242260186 Ack=2337482670 Win=64079 Len=0
15 1.778021	10.1.10.20	10.1.10.130	CIP PC	127	PCCC Class - Execute PCCC
16 1.788189	10.1.10.130	10.1.10.20	CIP PC	117	Success: PCCC Class - Execute PCCC
17 1.981726	10.1.10.20	10.1.10.130	TCP	60	49242 → 44818 [ACK] Seq=1242260259 Ack=2337482733 Win=64016 Len=0
18 3.206293	10.1.10.20	10.1.10.130	CIP PC	127	PCCC Class - Execute PCCC
19 3.299247	10.1.10.130	10.1.10.20	CIP PC	145	Success: PCCC Class - Execute PCCC
20 3.498174	10.1.10.20	10.1.10.130	TCP	60	49242 → 44818 [ACK] Seq=1242260332 Ack=2337482824 Win=63925 Len=0
21 4.801504	10.1.10.20	10.1.10.130	CIP PC	127	PCCC Class - Execute PCCC
22 4.812100	10.1.10.130	10.1.10.20	CIP PC	127	Success: PCCC Class - Execute PCCC
23 5.009981	10.1.10.20	10.1.10.130	TCP	60	49242 → 44818 [ACK] Seq=1242260405 Ack=2337482897 Win=63852 Len=0
24 6.333303	10.1.10.20	10.1.10.130	CIP PC	127	PCCC Class - Execute PCCC
25 6.335277	10.1.10.130	10.1.10.20	CIP PC	117	Success: PCCC Class - Execute PCCC
26 6.536142	10.1.10.20	10.1.10.130	TCP	60	49242 → 44818 [ACK] Seq=1242260478 Ack=2337482960 Win=63789 Len=0
27 7.859559	10.1.10.20	10.1.10.130	CIP PC	127	PCCC Class - Execute PCCC
28 7.859033	10.1.10.130	10.1.10.20	CIP PC	117	Success: PCCC Class - Execute PCCC
29 8.060800	10.1.10.20	10.1.10.130	TCP	60	49242 → 44818 [ACK] Seq=1242260551 Ack=2337483023 Win=63726 Len=0
30 9.365471	10.1.10.20	10.1.10.130	CIP PC	127	PCCC Class - Execute PCCC
31 9.370632	10.1.10.130	10.1.10.20	CIP PC	145	Success: PCCC Class - Execute PCCC
32 9.570136	10.1.10.20	10.1.10.130	TCP	60	49242 → 44818 [ACK] Seq=1242260624 Ack=2337483114 Win=63635 Len=0
33 10.378054	10.1.10.20	10.1.10.130	CIP PC	127	PCCC Class - Execute PCCC
34 10.889911	10.1.10.130	10.1.10.20	CIP PC	127	Success: PCCC Class - Execute PCCC
35 11.089102	10.1.10.20	10.1.10.130	TCP	60	49242 → 44818 [ACK] Seq=1242260697 Ack=2337483187 Win=63562 Len=0
36 12.395819	10.1.10.20	10.1.10.130	CIP PC	127	PCCC Class - Execute PCCC
37 12.401749	10.1.10.130	10.1.10.20	CIP PC	117	Success: PCCC Class - Execute PCCC
38 12.602365	10.1.10.20	10.1.10.130	TCP	60	49242 → 44818 [ACK] Seq=1242260770 Ack=2337483250 Win=63499 Len=0
39 13.906784	10.1.10.20	10.1.10.130	CIP PC	127	PCCC Class - Execute PCCC
40 13.915542	10.1.10.130	10.1.10.20	CIP PC	117	Success: PCCC Class - Execute PCCC
41 14.130288	10.1.10.20	10.1.10.130	TCP	60	49242 → 44818 [ACK] Seq=1242260843 Ack=2337483313 Win=63436 Len=0
42 15.425417	10.1.10.20	10.1.10.130	CIP PC	129	PCCC Class - Execute PCCC
43 15.435478	10.1.10.130	10.1.10.20	CIP PC	115	Success: PCCC Class - Execute PCCC
44 15.435488	10.1.10.20	10.1.10.130	CIP PC	127	PCCC Class - Execute PCCC

Using “HMI2PLC.pcap”, I can observe that .20 initiates communication with .130 and continues to send requests at regular intervals. The packet data shows .20 sending "CIP PCCC Class – Execute PCCC" commands, which are promptly acknowledged by .130. This pattern suggests that .20 consistently requests data at specific intervals, while the .130 device responds accordingly. This is best described as: **the .20 device requests data at regular intervals.**

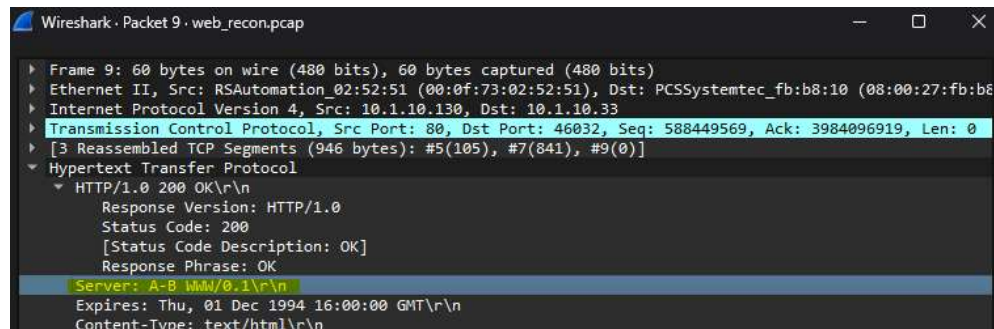
Question 14: What username was successfully used to access pages on the webserver on .130?

http contains "username"					
Time	Source	Destination	Protocol	Length	Info
243 23.951688	10.1.10.33	10.1.10.130	HTTP	731	GET /dataview.htm HTTP/1.1
264 24.163018	10.1.10.33	10.1.10.130	HTTP	687	GET /radevice.css HTTP/1.1
280 24.243619	10.1.10.33	10.1.10.130	HTTP	666	GET /URLhdl.js HTTP/1.1
292 24.303511	10.1.10.33	10.1.10.130	HTTP	672	GET /TableList.js HTTP/1.1

```
[...]Authorization: Digest username="guest",
\r\n
```

Using “web_recon.pcap”, I was able to filter for relevant HTTP packets containing “username”. I selected one of the packets and found that the username used to access the web server was **guest**.

Question 17: What webserver appears to be running on the .130 device?

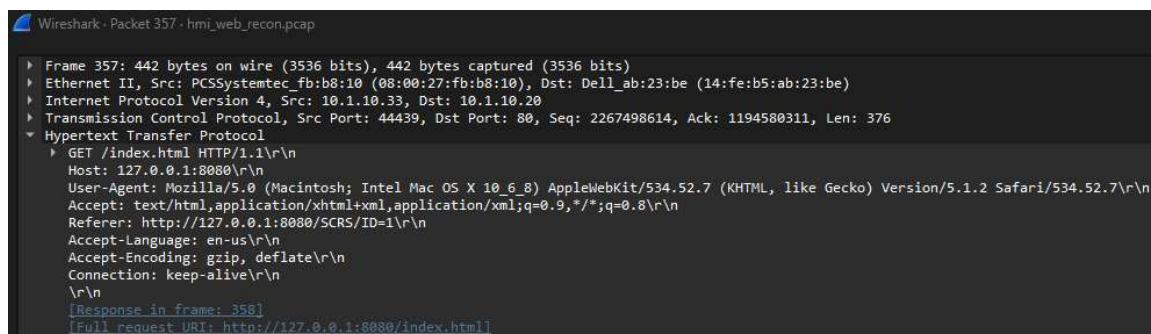


```
Wireshark · Packet 9 · web_recon.pcap
▶ Frame 9: 60 bytes on wire (480 bits), 60 bytes captured (480 bits)
▶ Ethernet II, Src: RSAutomation_02:52:51 (00:0f:73:02:52:51), Dst: PCSSystemtec_fb:b8:10 (08:00:27:fb:b8:10)
▶ Internet Protocol Version 4, Src: 10.1.10.130, Dst: 10.1.10.33
▶ Transmission Control Protocol, Src Port: 80, Dst Port: 46032, Seq: 588449569, Ack: 3984096919, Len: 0
▶ [3 Reassembled TCP Segments (946 bytes): #5(105), #7(841), #9(0)]
▼ Hypertext Transfer Protocol
  HTTP/1.0 200 OK\r\n
    Response Version: HTTP/1.0
    Status Code: 200
    [Status Code Description: OK]
    Response Phrase: OK
    Server: A-B WWW/0.1\r\n
    Expires: Thu, 01 Dec 1994 16:00:00 GMT\r\n
    Content-Type: text/html\r\n
```

Using “web_recon.pcap”, and selecting one of the HTTP OK packets, I was able to find that the webserver running on .130 was **A-B WWW/0.1**.

Question 19: What browser did the attackers use to access the HMI?

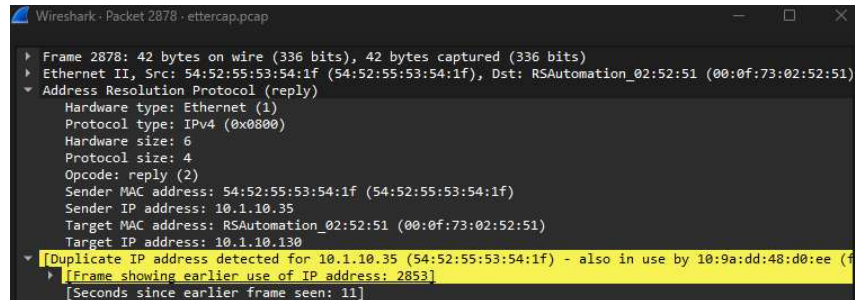
http.user_agent						
No.	Time	Source	Destination	Protocol	Length	Info
4	0.061350	10.1.10.33	10.1.10.20	HTTP	390	GET / HTTP/1.1
14	9.617898	10.1.10.33	10.1.10.20	HTTP	657	GET / HTTP/1.1
24	73.213589	10.1.10.33	10.1.10.20	HTTP	432	GET /SCRL.html HTTP/1.1



```
Wireshark · Packet 357 · hmi_web_recon.pcap
▶ Frame 357: 442 bytes on wire (3536 bits), 442 bytes captured (3536 bits)
▶ Ethernet II, Src: PCSSystemtec_fb:b8:10 (08:00:27:fb:b8:10), Dst: Dell_ab:23:be (14:fe:b5:ab:23:be)
▶ Internet Protocol Version 4, Src: 10.1.10.33, Dst: 10.1.10.20
▶ Transmission Control Protocol, Src Port: 44439, Dst Port: 80, Seq: 2267498614, Ack: 1194580311, Len: 376
▼ Hypertext Transfer Protocol
  GET /index.html HTTP/1.1\r\n
    Host: 127.0.0.1:8080\r\n
    User-Agent: Mozilla/5.0 (Macintosh; Intel Mac OS X 10_6_8) AppleWebKit/534.52.7 (KHTML, like Gecko) Version/5.1.2 Safari/534.52.7\r\n
    Accept: text/html,application/xhtml+xml,application/xml;q=0.9,*/*;q=0.8\r\n
    Referer: http://127.0.0.1:8080/SCRS/ID=1\r\n
    Accept-Language: en-us\r\n
    Accept-Encoding: gzip, deflate\r\n
    Connection: keep-alive\r\n
    \r\n
    [Response in frame: 358]
    [Full request URI: http://127.0.0.1:8080/index.html]
```

Using “hmi_web_recon.pcap” with the filter “http.user_agent”, I determined that the attackers used the **Safari** browser to access the HMI. This filter allowed me to quickly locate packets containing the “User-Agent” field, which provides browser details. In the packets I reviewed, the “User-Agent” field shows “... Version/5.1.2 Safari...” confirming Safari version 5.1.2 on MacOS.

Question 27: What appears to be the MAC address of the device which performed the ARP spoofing?



Using “ettercap.pcap” and filter “arp && eth.addr == 00:0f:73:02:52:51 && not eth.dst == ff:ff:ff:ff:ff:ff”, I was able to narrow down my options. This meant I could now see only ARP packets related to one of the given MAC addresses and I wanted to ignore all the Broadcast destination packets. I selected a couple of packets to inspect more closely. It didn’t take long for me to find the “Duplicate IP address ...” message, confirming **00:0f:73:02:52:51** was the MAC address of the device which performed the ARP spoofing.

Summary

In this lab, I used Wireshark to analyze six packet capture files and answer questions about the captured network activity. I applied filters and looked closely at packet details to find open ports, services, software versions, and unusual communication patterns. This helped me learn how to investigate network traffic and find important information about devices and connections. While I am still getting used to Wireshark, this lab gave me good practice with filters and basic network forensics.